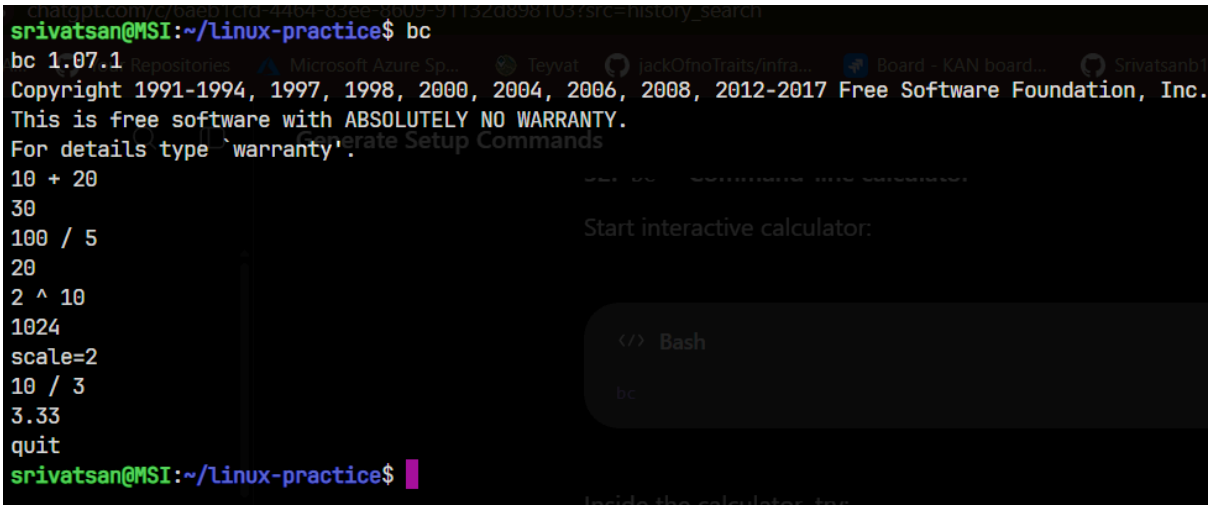
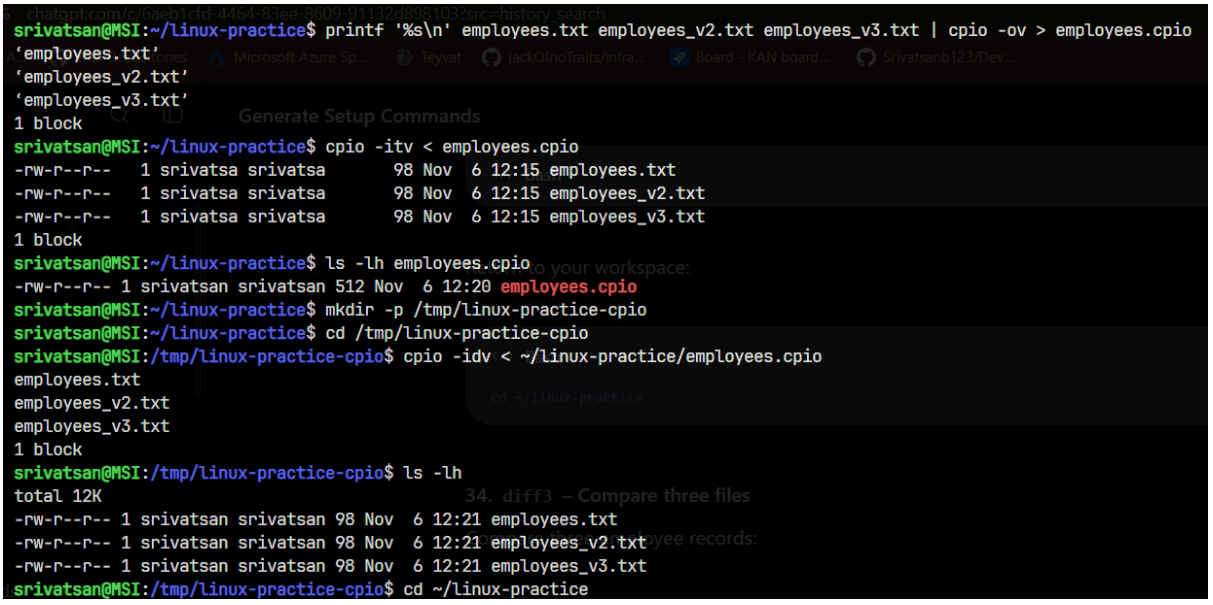
bc:



```
srivatsan@MSI:~/linux-practice$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
10 + 20
30
100 / 5
20
2 ^ 10
1024
scale=2
10 / 3
3.33
quit
srivatsan@MSI:~/linux-practice$
```

cpio:



```
srivatsan@MSI:~/linux-practice$ printf '%s\n' employees.txt employees_v2.txt employees_v3.txt | cpio -ov > employees.cpio
'employees.txt'
'employees_v2.txt'
'employees_v3.txt'
1 block
srivatsan@MSI:~/linux-practice$ cpio -itv < employees.cpio
-rw-r--r--   1 srivatsa srivatsa       98 Nov  6 12:15 employees.txt
-rw-r--r--   1 srivatsa srivatsa       98 Nov  6 12:15 employees_v2.txt
-rw-r--r--   1 srivatsa srivatsa       98 Nov  6 12:15 employees_v3.txt
1 block
srivatsan@MSI:~/linux-practice$ ls -lh employees.cpio
-rw-r--r-- 1 srivatsan srivatsan 512 Nov  6 12:20 employees.cpio
srivatsan@MSI:~/linux-practice$ mkdir -p /tmp/linux-practice-cpio
srivatsan@MSI:~/linux-practice$ cd /tmp/linux-practice-cpio
srivatsan@MSI:/tmp/linux-practice-cpio$ cpio -idv < ~/linux-practice/employees.cpio
employees.txt
employees_v2.txt
employees_v3.txt
1 block
srivatsan@MSI:/tmp/linux-practice-cpio$ ls -lh
total 12K
-rw-r--r-- 1 srivatsan srivatsan 98 Nov  6 12:21 employees.txt
-rw-r--r-- 1 srivatsan srivatsan 98 Nov  6 12:21 employees_v2.txt
-rw-r--r-- 1 srivatsan srivatsan 98 Nov  6 12:21 employees_v3.txt
srivatsan@MSI:/tmp/linux-practice-cpio$ cd ~/linux-practice
```

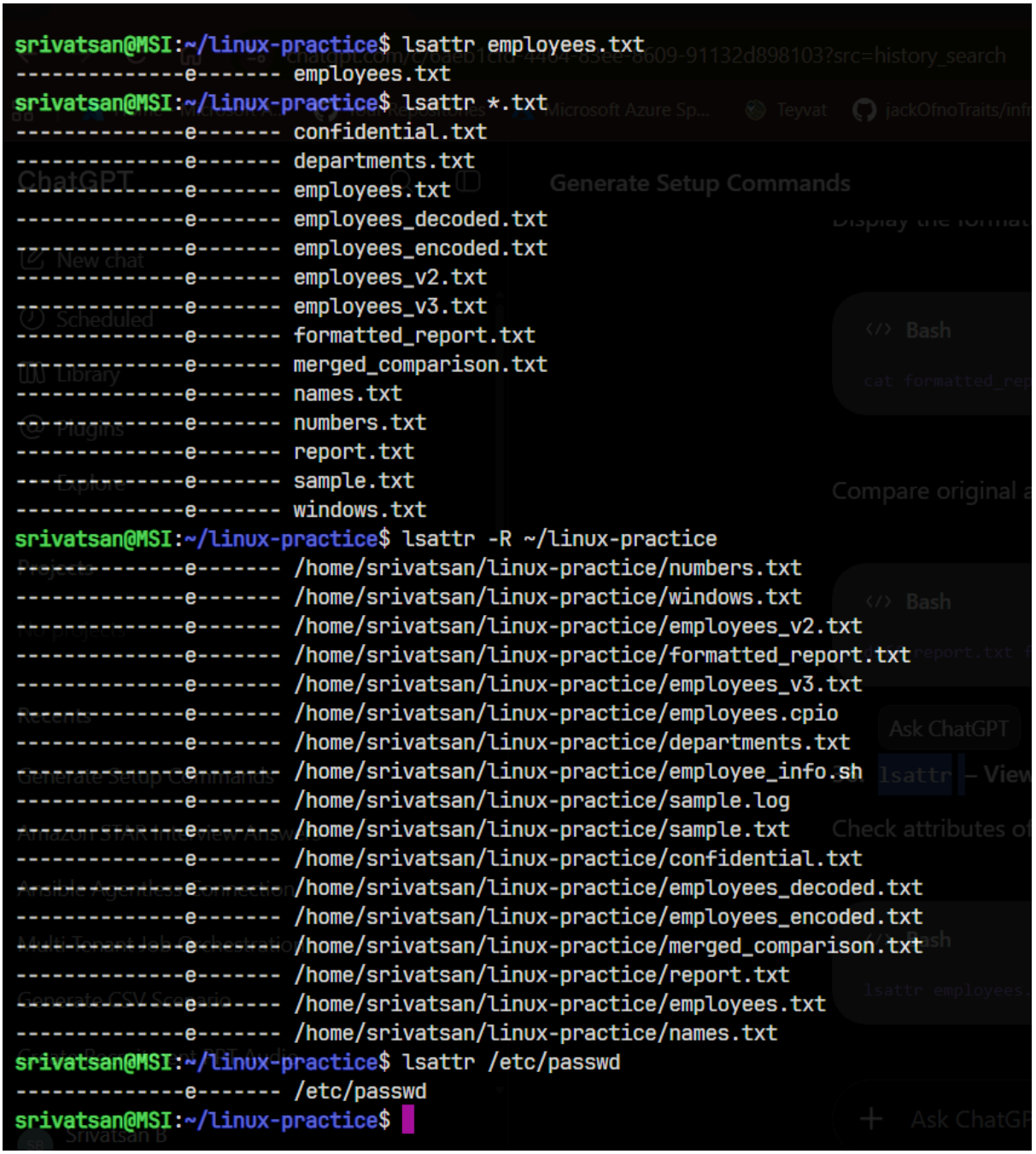
diff:

```
srivatsan@MSI:~/linux-practice$ diff3 -x employees.txt employees_v2.txt employees_v3.txt
1,5c
101 Alice HR 25000
102 Bob IT 36000
103 Charlie Finance 30000
104 David HR 29000
105 Eva IT 40000
.
srivatsan@MSI:~/linux-practice$ diff3 -m employees.txt employees_v2.txt employees_v3.txt > merged_comparison.txt
srivatsan@MSI:~/linux-practice$ cat merged_comparison.txt
<<<<<<< employees.txt
101 Alice HR 25000
102 Bob IT 35000
103 Charlie Finance 30000
104 David HR 28000
105 Eva IT 40000
||||||| employees_v2.txt
101 Alice HR 27000
102 Bob IT 35000
103 Charlie Finance 32000
104 David HR 28000
105 Eva IT 42000
=======
101 Alice HR 25000
102 Bob IT 36000
103 Charlie Finance 30000
104 David HR 29000
105 Eva IT 40000
>>>>>>> employees_v3.txt
srivatsan@MSI:~/linux-practice$
```

fmt:

```
srivatsan@MSI:~/linux-practice$ cat report.txt
Linux administration involves managing files, processes, permissions, storage, and system resources.
Command line utilities help administrators automate repetitive operations and troubleshoot system problems.
Learning these commands improves our understanding of Linux operating systems.
srivatsan@MSI:~/linux-practice$ fmt -w 40 report.txt
Linux administration involves managing
files, processes, permissions, storage,
and system resources.  Command line
utilities help administrators automate
repetitive operations and troubleshoot
system problems.  Learning these commands
improves our understanding of Linux
operating systems.
srivatsan@MSI:~/linux-practice$ fmt -w 60 report.txt
Linux administration involves managing files, processes,
permissions, storage, and system resources.  Command line
utilities help administrators automate repetitive operations
and troubleshoot system problems.  Learning these commands
improves our understanding of Linux operating systems.
srivatsan@MSI:~/linux-practice$ fmt -w 50 report.txt > formatted_report.txt
srivatsan@MSI:~/linux-practice$ cat formatted_report.txt
Linux administration involves managing files,
processes, permissions, storage, and system
resources.  Command line utilities help
administrators automate repetitive operations
and troubleshoot system problems.  Learning
these commands improves our understanding of
Linux operating systems.
srivatsan@MSI:~/linux-practice$
```

lsattr:


srivatsan@MSI:~/linux-practice$ lsattr employees.txt
----------e------- employees.txt
srivatsan@MSI:~/linux-practice$ lsattr *.txt
----------e------- confidential.txt
----------e------- departments.txt
----------e------- employees.txt
----------e------- employees_decoded.txt
----------e------- employees_encoded.txt
----------e------- employees_v2.txt
----------e------- employees_v3.txt
----------e------- formatted_report.txt
----------e------- merged_comparison.txt
----------e------- names.txt
----------e------- numbers.txt
----------e------- report.txt
----------e------- sample.txt
----------e------- windows.txt
srivatsan@MSI:~/linux-practice$ lsattr -R ~/linux-practice
----------e------- /home/srivatsan/linux-practice/numbers.txt
----------e------- /home/srivatsan/linux-practice/windows.txt
----------e------- /home/srivatsan/linux-practice/employees_v2.txt
----------e------- /home/srivatsan/linux-practice/formatted_report.txt
----------e------- /home/srivatsan/linux-practice/employees_v3.txt
----------e------- /home/srivatsan/linux-practice/employees.cpio
----------e------- /home/srivatsan/linux-practice/departments.txt
----------e------- /home/srivatsan/linux-practice/employee_info.sh
----------e------- /home/srivatsan/linux-practice/sample.log
----------e------- /home/srivatsan/linux-practice/sample.txt
----------e------- /home/srivatsan/linux-practice/confidential.txt
----------e------- /home/srivatsan/linux-practice/employees_decoded.txt
----------e------- /home/srivatsan/linux-practice/employees_encoded.txt
----------e------- /home/srivatsan/linux-practice/merged_comparison.txt
----------e------- /home/srivatsan/linux-practice/report.txt
----------e------- /home/srivatsan/linux-practice/employees.txt
----------e------- /home/srivatsan/linux-practice/names.txt
srivatsan@MSI:~/linux-practice$ lsattr /etc/passwd
----------e------- /etc/passwd
srivatsan@MSI:~/linux-practice$

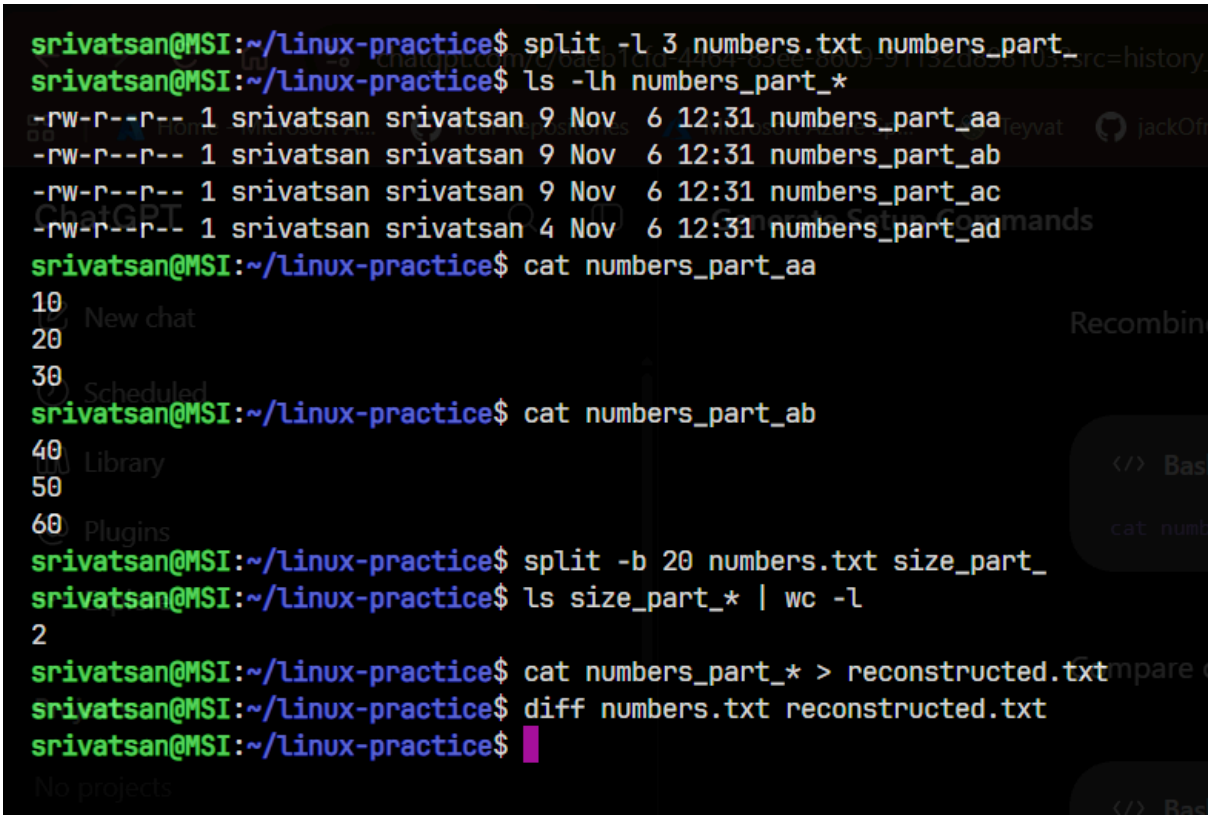
lsof:

```
srivatsan@MSI:~/linux-practice$ exec 3< employees.txt
srivatsan@MSI:~/linux-practice$ lsof employees.txt
COMMAND PID      USER FD   TYPE DEVICE SIZE/OFF NODE NAME
bash    464 srivatsan 3r   REG   8,48       98 2091 employees.txt
srivatsan@MSI:~/linux-practice$ lsof employees.txt
COMMAND PID      USER FD   TYPE DEVICE SIZE/OFF NODE NAME
bash    464 srivatsan 3r   REG   8,48       98 2091 employees.txt
srivatsan@MSI:~/linux-practice$ lsof +D ~/linux-practice
COMMAND   PID     USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
bash      464 srivatsan cwd    DIR   8,48     4096 2485 /home/srivatsan/linux-practice
bash      464 srivatsan   3r   REG   8,48       98 2091 /home/srivatsan/linux-practice/employees.txt
lsof     1359 srivatsan cwd    DIR   8,48     4096 2485 /home/srivatsan/linux-practice
lsof     1360 srivatsan cwd    DIR   8,48     4096 2485 /home/srivatsan/linux-practice
srivatsan@MSI:~/linux-practice$ sudo lsof -i :22
srivatsan@MSI:~/linux-practice$ exec 3<&-
srivatsan@MSI:~/linux-practice$
```

nice:

```
srivatsan@MSI:~/linux-practice$ nice -n 10 sleep 60 &
[1] 1422
srivatsan@MSI:~/linux-practice$ PID=$!
srivatsan@MSI:~/linux-practice$ ps -o pid,ni,comm -p "$PID"
    PID  NI COMMAND
   1422  10 sleep
srivatsan@MSI:~/linux-practice$ nice -n 15 sleep 60 &
[2] 1453
srivatsan@MSI:~/linux-practice$ ps -eo pid,ni,comm | grep sleep
   1422  10 sleep
   1453  15 sleep
srivatsan@MSI:~/linux-practice$ pkill -x sleep
[1]-  Terminated                 nice -n 10 sleep 60
[2]+  Terminated                 nice -n 15 sleep 60
srivatsan@MSI:~/linux-practice$
```

split:


srivatsan@MSI:~/linux-practice$ split -l 3 numbers.txt numbers_part_
srivatsan@MSI:~/linux-practice$ ls -lh numbers_part_*
-rw-r--r-- 1 srivatsan srivatsan 9 Nov 6 12:31 numbers_part_aa
-rw-r--r-- 1 srivatsan srivatsan 9 Nov 6 12:31 numbers_part_ab
-rw-r--r-- 1 srivatsan srivatsan 9 Nov 6 12:31 numbers_part_ac
-rw-r--r-- 1 srivatsan srivatsan 4 Nov 6 12:31 numbers_part_ad
srivatsan@MSI:~/linux-practice$ cat numbers_part_aa
10
20
30
srivatsan@MSI:~/linux-practice$ cat numbers_part_ab
40
50
60
srivatsan@MSI:~/linux-practice$ split -b 20 numbers.txt size_part_
srivatsan@MSI:~/linux-practice$ ls size_part_* | wc -l
2
srivatsan@MSI:~/linux-practice$ cat numbers_part_* > reconstructed.txt
srivatsan@MSI:~/linux-practice$ diff numbers.txt reconstructed.txt
srivatsan@MSI:~/linux-practice$

stat:

```
srivatsan@MSI:~/linux-practice$ stat employees.txt
  File: employees.txt
  Size: 98              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 2091        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/srivatsan)   Gid: ( 1000/srivatsan)
Access: 2026-11-06 12:20:42.646934453 +0000
Modify: 2026-11-06 12:15:38.620786887 +0000
Change: 2026-11-06 12:15:38.620786887 +0000
 Birth: 2026-11-05 12:54:44.337098104 +0000
srivatsan@MSI:~/linux-practice$ stat -c "%s bytes" employees.txt
98 bytes
srivatsan@MSI:~/linux-practice$ stat -c "%A %a" employees.txt
-rw-r--r-- 644
srivatsan@MSI:~/linux-practice$ stat -c "%U %G" employees.txt
srivatsan srivatsan
srivatsan@MSI:~/linux-practice$ stat -c "%y" employees.txt
2026-11-06 12:15:38.620786887 +0000
srivatsan@MSI:~/linux-practice$ stat -c "%i" employees.txt
2091
srivatsan@MSI:~/linux-practice$ stat -f ~/linux-practice
  File: "/home/srivatsan/linux-practice"
    ID: 9cbda8d3e037c4a6 Namelen: 255     Type: ext2/ext3
Block Size: 4096       Fundamental block size: 4096
Blocks: Total: 263940717  Free: 263507480  Available: 250081612
Inodes: Total: 67108864   Free: 67062873
srivatsan@MSI:~/linux-practice$
```

xxd:

```
srivatsan@MSI:~/linux-practice$ printf 'Linux Practice 2026\n' > binary_sample.bin
srivatsan@MSI:~/linux-practice$ xxd binary_sample.bin
00000000: 4c69 6e75 7820 5072 6163 7469 6365 2032  Linux Practice 2
00000010: 3032 360a                                026.
srivatsan@MSI:~/linux-practice$ xxd -p binary_sample.bin
4c696e7578205072616374696365203230323630a
srivatsan@MSI:~/linux-practice$ xxd -p binary_sample.bin
4c696e7578205072616374696365203230323630a
srivatsan@MSI:~/linux-practice$ xxd -l 8 binary_sample.bin
00000000: 4c69 6e75 7820 5072                      Linux Pr
srivatsan@MSI:~/linux-practice$ xxd binary_sample.bin > binary_dump.txt
srivatsan@MSI:~/linux-practice$ xxd -r binary_dump.txt > restored.bin
srivatsan@MSI:~/linux-practice$ cmp binary_sample.bin restored.bin
srivatsan@MSI:~/linux-practice$ xxd -p binary_sample.bin
4c696e7578205072616374696365203230323630a
srivatsan@MSI:~/linux-practice$
```